Reliability-weighted target prioritization in CD4+ T-cell Perturb-seq: a generalizability-theory decomposition

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Abstract

Genome-scale Perturb-seq screens prioritize candidate targets by the strength of a perturbation's transcriptional effect. Effect strength does not answer a prior measurement question: is the readout dependable? A large effect estimated from a single guide, a single donor, or a pseudobulk of few cells need not survive replication, and for target prioritization each false lead costs a validation experiment. We treat each perturbation effect as a measurement in a crossed Target × Guide × Donor × Condition design and apply generalizability theory (Cronbach et al., 1972) to separate the dependable part of an effect from facet-specific idiosyncrasy. Guides and donors enter as random facets; condition enters as a fixed facet and is analyzed within its levels. For each target we report a dependability profile over the facets and a joint generalizability coefficient over the two random facets, and we re-rank targets by effect magnitude weighted by that coefficient. On the released screen (Zhu et al., 2025), removing the measurement-error floor estimated from the non-targeting controls raises the number of genes with a dependable target-signal share above .10 from 40 to 7,674. A design study indicates that reliability is limited by the number of guides rather than the number of donors. Analyzed within activation states, dependability recovers the T-cell-receptor signaling module as reliably measurable only in activated cells, without recourse to gene annotation. Every methodological decision was recorded and adversarially reviewed, and all results regenerate from the released summary statistics.

1 Dependability in target prioritization

Genome-scale Perturb-seq couples pooled CRISPR perturbation with a single-cell transcriptomic readout, so that the transcriptional consequence of perturbing each of thousands of genes is measured in one experiment (Dixit et al., 2016; Replogle et al., 2022). Applied to primary human CD4+ T cells, the cells that coordinate adaptive immunity and are a substrate for engineered cell therapies, such a screen identifies the genes whose perturbation most reshapes a T-cell program and, on that basis, nominates candidate targets (Zhu et al., 2025). The nomination carries a cost. Each target that advances is committed to an arrayed knockdown, a functional assay, or an animal model, so a hit that does not reproduce spends a validation experiment on an artifact. What a screen is worth therefore depends not only on the effects it detects but on whether the ranking it induces over targets can be trusted.

Target prioritization conventionally ranks candidates by the strength of a perturbation's differential-expression signature. The strength is read from the coefficients of a regularized model of expression on guide assignment (Dixit et al., 2016), from a z-score or false-discovery rate for each perturbation-gene association, or from the number of downstream differentially expressed genes together with a global test that the transcriptome moved at all (Replogle et al., 2022). Effect metrics have since grown more sophisticated, replacing the gene count with the distance between the perturbed and control single-cell distributions (Peidli et al., 2024) or with a per-target rate of cells classified as perturbed (Papalexi et al., 2021). All of them quantify how large an effect is.

None answers a prior question: whether the effect is dependable, in the sense that it would recur under another guide, another donor, or another cell-state context. The number of differentially expressed genes, for instance, tracks the number of cells assayed (Dixit et al., 2016) and need not track the strength of the perturbation itself (Norman et al., 2019), so a target can rank high on this measure yet low on reproducibility. A large apparent effect can be fragile, when two guides disagree, one donor drives the signal, knockdown is weak, or an off-target flag is present, while a target of moderate but consistent effect is the better validation bet. We therefore treat *dependability* as a first-class quantity alongside magnitude.

Two properties are at stake that effect size does not by itself deliver: *accuracy*, that a reported effect is dependable rather than measurement noise, and *generalizability*, that it recurs across guides, donors, and cell-state contexts. We therefore ask four questions, moving from the present screen to future ones. *RQ1 (per target):* which targets carry an individually dependable effect, one that would recur under a fresh guide and a fresh donor, and does ranking by that dependability reorder the list that effect magnitude alone produces? *RQ2 (in context):* does the same dependability, evaluated within each activation state, separate context-specific signal from noise and recover coherent biology? *RQ3 (the design):* how dependable is the present screen as a whole, at its two guides and four donors? *RQ4 (improving the design):* which facet limits that dependability, and how much would a future screen recover by adding to it? RQ1 and RQ2 concern individual targets, RQ3 and RQ4 the measurement design; all four presuppose that the dependable part of an effect can first be separated from measurement noise.

Reliability, as opposed to size, has been approached from two directions. One measures reproducibility across replicates directly: the irreproducible discovery rate calls discoveries by whether they recur rather than by effect size (Li et al., 2011), and reproducibility metrics tuned to context-specific CRISPR screens show that standard quality-control does not capture whether such a hit will replicate (Billmann et al., 2023). The other decomposes expression variance into named sources, as a per-gene linear mixed model partitions variance into donor, batch, and residual components (Hoffman & Schadt, 2016); data-driven prioritization frameworks likewise integrate several weighted evidence types rather than fitness effect alone (Behan et al., 2019). Neither direction yields a per-target coefficient one can rank on. Generalizability theory joins them: it turns a per-gene variance decomposition into a single dependability coefficient for each target, on the scale of a correlation between independent repeats, and separates the facets a target must generalize across from the context it is measured within.

The rest of the paper develops the per-target coefficient and the two levels it rests on (Section 2), describes the data and the pre-registered choice of estimator (Section 3), and takes up RQ1 through RQ4 in the results (Section 4).

2 A generalizability-theory decomposition

2.1 The crossed design and its facets

We treat one perturbation's effect as a measurement and ask how much of it would recur if the experiment were repeated. The quantity of interest is the target gene's true effect on the transcriptome. What we observe is that effect seen through one guide, in one donor, under one culture condition, and read from a finite sample of cells, and each of these can make the observation differ from the truth. This is the setting generalizability theory was built for, and the correspondence is direct (Lord & Novick, 1968): the target gene is the *object of measurement*, its true effect is the *universe score* (the value one would obtain by averaging over infinitely many independent repeats), the two guides are interchangeable versions of the same perturbation (*parallel forms*), and donors and conditions are further experimental factors, or *facets*, crossed with the target. Cell sampling and sequencing contribute the residual noise. Classical test theory writes an observation as a true score plus error, $X = T + E$ (Novick, 1966), and generalizability theory (Cronbach et al., 1972) splits that error into one term per facet,

$$E = E_{\text{guide}} + E_{\text{donor}} + E_{\text{condition}} + E_{\text{interactions}} + E_{\text{residual}},$$

and estimates how large each term is.

We distinguish random from fixed facets. Guides and donors are *random* facets: one wants the effect to generalize across the guides and donors that were not sampled. Condition is a *fixed* facet: Rest, Stim8hr, and Stim48hr are three deliberately chosen activation states, not a sample from a population of conditions, so the appropriate treatment is a separate analysis within each state rather than a generalization across states. A perturbation that acts only in the activated state is context-specific, not unreliable, and folding condition into a generalization coefficient would misclassify that signal as noise.

2.2 Per-target dependability

For target t we report dependability as a *profile* of three coefficients, one per facet, kept separate and never multiplied together. Each coefficient answers a reproducibility question: with everything held fixed except one facet, how well does the target's transcriptional signature repeat? An observation is a (target, guide, donor, condition) pseudobulk whose effect profile spans the 18,129 genes, measured relative to a matched non-targeting control. We average a group of observations into a mean profile, and each facet coefficient is the Pearson correlation, across genes, between two mean profiles that differ only in that facet:

- R_t^{guide}: the target's guides are split into two halves, each half's mean profile is formed, and the two are correlated. A high value indicates that the transcriptional signature reproduces across independent guide sets.[1]
- R_t^{donor}: the same construction with a two-versus-two donor split, then shrunk toward the pooled mean by a moment-based empirical-Bayes step, which stabilizes a coefficient that four donors otherwise leave with only three degrees of freedom.

[1] These are raw single-observation split-half correlations. The reliability of the deployed two-guide average is the Spearman–Brown step-up $2r/(1+r)$, a separate and larger quantity; we report the single-observation form because the joint coefficient is defined at the single-guide-by-single-donor level. Both are distribution-light empirical correlations rather than Gaussian variance-component ratios.

- $R_t^{\text{condition}}$: the mean of the pairwise correlations among the three within-state profiles, reported as cross-context *consistency* rather than as a generalization coefficient.

To rank targets we need one number rather than a profile. The per-target dependability $R_{\text{dep},t}$ is that number, and it has a concrete meaning: the correlation one would expect between two fully independent repeats of the experiment, a second run carried out with fresh guides and fresh donors. Writing an observed profile as a true signal plus independent guide and donor errors, $\tau_t + \varepsilon_{\text{guide}} + \varepsilon_{\text{donor}}$, that correlation is the signal's share of the total variation, $\sigma_\tau^2/(\sigma_\tau^2 + \sigma_{\text{guide}}^2 + \sigma_{\text{donor}}^2)$. The same quantity is assembled from the two split-half coefficients. Each is itself a signal share of the same kind, $R = \sigma_\tau^2/(\sigma_\tau^2 + \sigma_e^2)$, where σ_e^2 is the error variance of the facet it refreshes. Its error-to-signal ratio is therefore $(1 - R)/R = \sigma_e^2/\sigma_\tau^2$, and because the guide and donor errors are independent these two ratios add:

$$R_{\text{dep},t} = \frac{1}{1 + \dfrac{1 - R_t^{\text{guide}}}{R_t^{\text{guide}}} + \dfrac{1 - R_t^{\text{donor}}}{R_t^{\text{donor}}}}, \qquad S_t = M_t \, R_{\text{dep},t}, \tag{1}$$

where M_t is the effect magnitude and S_t the dependability-weighted score used for ranking. A single-facet coefficient refreshes one source of variation; the joint coefficient refreshes both, and is therefore the most demanding form of reproducibility we report. The equality is exact when the two errors are independent, and becomes an upper bound if a guide-by-donor interaction adds a further disagreement term.

We keep a naming discipline that separates the two levels of the theory: $R_{\text{dep},t}$ is a per-target quantity, whereas the design-level coefficients $E\rho^2$ and Φ describe the measurement design as a whole and carry no per-target subscript.

3 Data and estimation

The data are the released summary statistics of a genome-scale Perturb-seq screen in primary human CD4+ T cells (Zhu et al., 2025): CRISPR perturbations of 18,129 genes in a crossed design of two guides per gene, four donors, and three culture conditions (Rest, Stim8hr, Stim48hr), with non-targeting guides as controls. We work from the released joint pseudobulk and define each perturbation's per-gene effect as its non-targeting-control-relative log-CPM within the matched donor-condition-run stratum, so that the effect is measured against the controls that share its technical context. No single-cell data are processed.

The variance contributed by each facet is estimated per gene by a crossed mixed model that weights each pseudobulk by its precision, following the generalizability-theory design of Section 2 (Brennan, 2001). We did not assume this estimator. Three candidates were compared on simulated data with known variance components, a pre-registered synthetic-recovery test, and only the precision-weighted model recovered the planted values under both Gaussian and heavy-tailed noise. A distance-based candidate failed, because a share of a distance is not a variance and cannot be read as one.

4 Results

4.1 Measurement error masks dependable signal

The residual of a pseudobulk decomposition combines biological idiosyncrasy with measurement error, and the latter is inflated when a pseudobulk aggregates few cells. The non-targeting controls

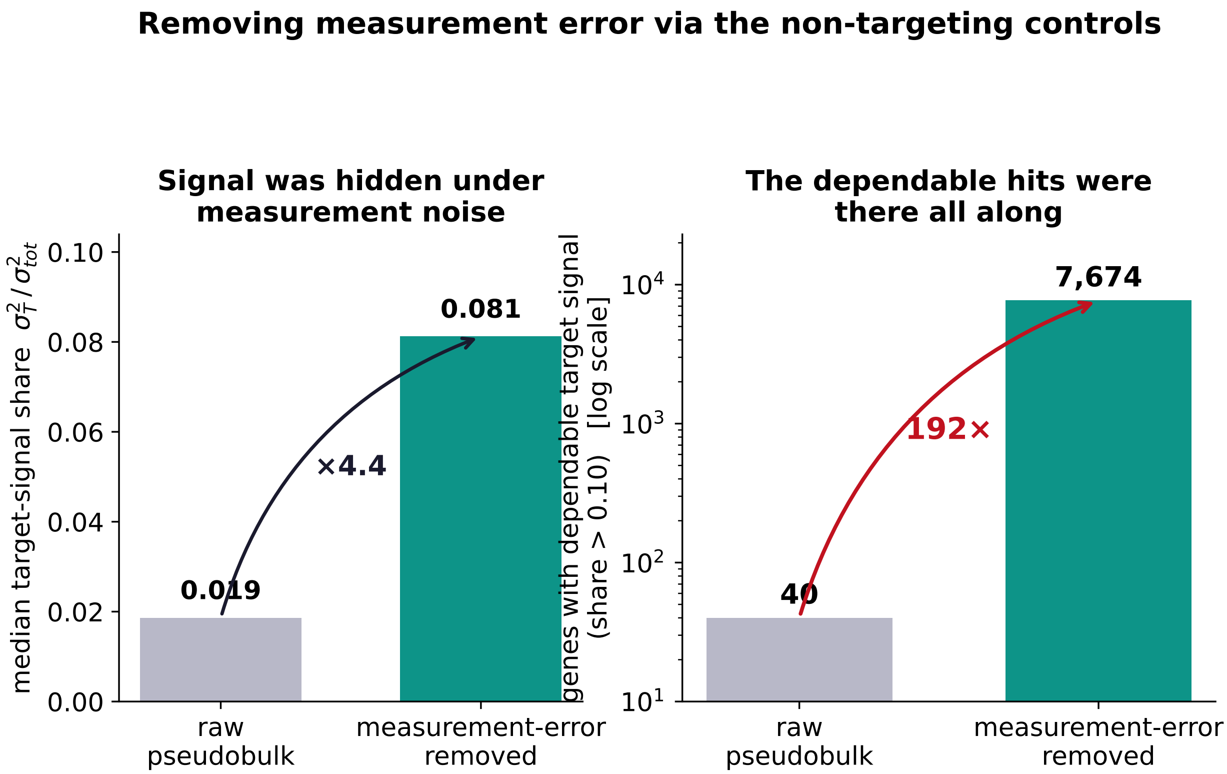

Figure 1: Measurement-error removal via the non-targeting controls. Left, the median target-signal share; right, the number of genes with a dependable target-signal share above .10, on a logarithmic axis.

carry no perturbation, so the variance of the non-targeting effect within its stratum estimates the per-gene measurement-error floor directly. Subtracting this floor from the residual raises the median target-signal share from .019 to .081 and raises the number of genes with a target-signal share above .10 from 40 to 7,674, or 42% of the genome (Figure 1). The dependable signal was present but masked; the same controls give a clean non-targeting-versus-non-targeting negative test ($r \approx 0$), so the correction introduces no signal where none exists.

4.2 A guide-limited design

A design study projects the generalizability coefficient onto alternative numbers of guides and donors (Figure 2). On the representative signal genes the current design (two guides, four donors) reaches $E\rho^2 \approx .44$. The guide error term exceeds the donor error term by a factor of about 2.4; adding one guide raises the coefficient about three times as much as adding one donor. With two guides the coefficient is capped near .53, and no number of donors reaches .70, whereas .70 is attainable with about 15 guides. Guide-to-guide variability, rather than donor heterogeneity, is the binding constraint, which is actionable guidance for the design of subsequent screens. Repeating the design study within each activation state returns the same verdict in all three states (Table 1), so the recommendation is not an artifact of pooling; within a state the guide term dominates the donor term more sharply still, because most donor disagreement is a cross-state phenomenon.

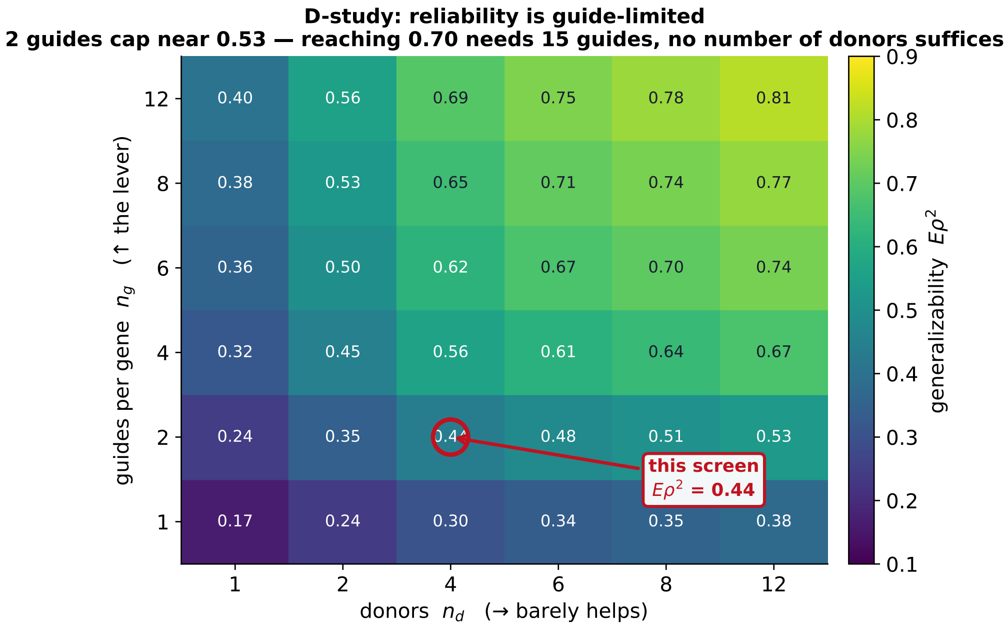

Figure 2: The generalizability coefficient $E\rho^2$ across the design grid. The current screen (two guides, four donors) is marked; reaching .70 requires about 15 guides and is unreachable by adding donors.

Table 1: Per-condition design study on the representative signal genes. The guide-limited verdict holds within every activation state.

Activation state	$E\rho^2$	Guide error	Donor error
Rest	.502	0.026	0.005
Stim8hr	.507	0.023	0.004
Stim48hr	.491	0.019	0.004
Pooled	.438	0.040	0.017

4.3 A reliability-weighted ranking

Ranking targets by S_t reorders the candidate list relative to effect size alone (Figure 3). Of the 100 highest-effect targets, 49 fall out of the 100 most dependable, and reproducible targets that effect size underranks rise (Table 2). These are the targets a dependability-weighted screen would nominate for validation but a strength-ranked screen would overlook.

4.4 Context-specific dependability recovers the TCR module

Analyzed within activation states, dependability separates targets that are reliable across all states from targets that are reliable in one state and not another (Figure 4). Because a within-state per-target dependability rests on roughly a third of the cells and runs lower than the pooled value, states are compared to one another rather than to the pooled number. One hundred targets are context-specific in this sense. Among them, the T-cell-receptor (TCR) signaling module, the CD3 complex (CD3D, CD3G, CD247) with the proximal kinase ZAP70 and the scaffold LAT, is reliably measurable only in activated cells, with resting dependability at the noise floor (resting $R_{\text{dep}} \approx .02$–$.05$; stimulated $\approx .20$–$.28$). The knockdown effect for CD3D grows with activation (effect

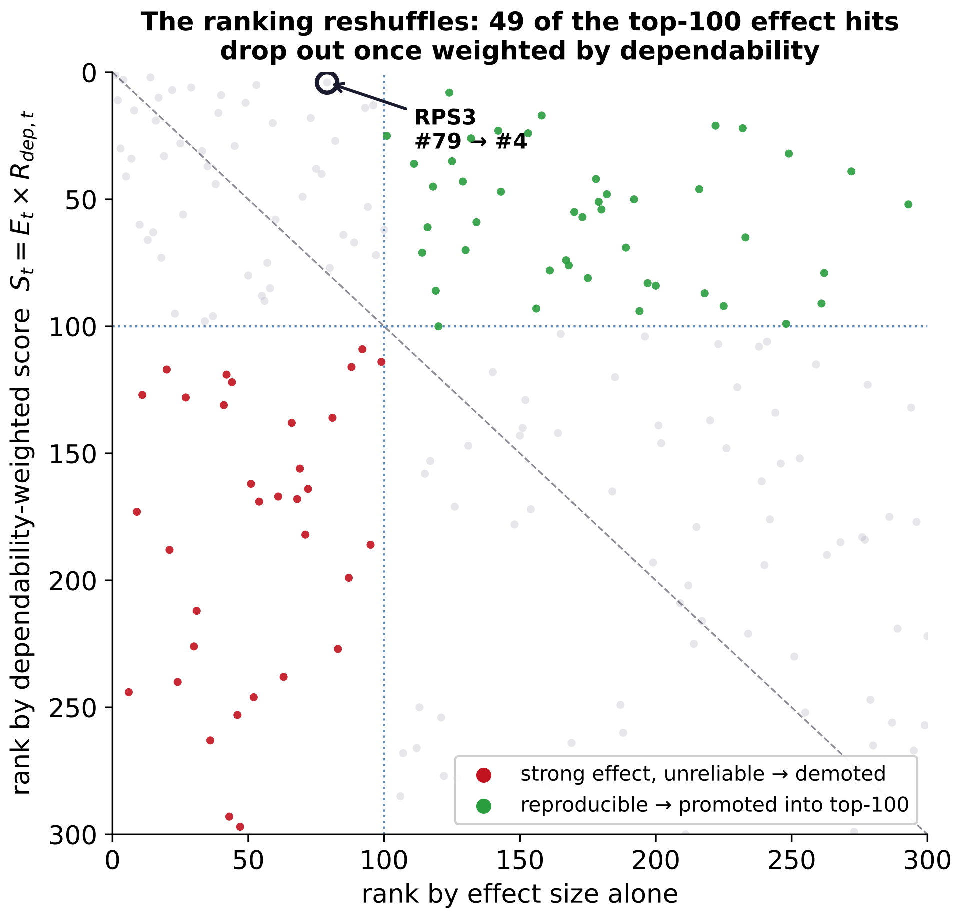

Figure 3: Rank by effect size alone (horizontal) against rank by the dependability-weighted score (vertical). Points below the diagonal in the lower-left quadrant are demoted (strong but unreliable); points in the upper right are promoted (reproducible).

magnitude 0.26, 0.57, and 1.12 across Rest, Stim8hr, and Stim48hr) and becomes dependable only upon activation. The framework reconstructs a coherent signaling module from dependability alone, without gene annotation, which is the strongest available evidence that it measures a real property of the perturbation rather than an artifact of the summary.

5 Reproducibility

The pipeline runs end to end from the released summary statistics, without processing the full single-cell data: data retrieval, a checksummed evidence manifest, identifiability gates, synthetic method selection, per-gene decomposition, measurement-error removal, the design study, and the per-context ranking. All results are committed as tables and figures, and every reported number regenerates from a single re-run. The code is released under Apache-2.0.

Table 2: Targets that effect size underranks but dependability promotes. Rank is out of the 7,169 scored targets.

Target	Rank by effect	Rank by dependability	$R_{\mathrm{dep},t}$
RPS3	79	4	.34
QARS1	53	5	.32
RPP21	124	8	.34
IMP4	96	13	.31
NMD3	158	17	.33

6 Limitations

Four donors leave the donor variance component with three degrees of freedom, so the donor coefficient and the donor axis of the design study carry real uncertainty, which we report as bands rather than point values. Sequencing run is partially confounded with donor, a fact of the released design that we record rather than correct. A small number of full-scale Monte Carlo gate confirmations are scheduled on a compute cluster; the results shown are the local synthetic and real-data tier. Finally, this is a re-analysis of released data, so the design-study guidance is prospective, addressed to subsequent screens rather than to the present one.

A A recorded, cross-model-hardened process

The methodology was developed under an auditable process (Figure 5). Each decision (the choice of estimator, the thresholds, the classification of each facet as random or fixed, and the criterion for identifiability) was recorded as a versioned issue using issue-driven development (Cheng, 2026a) and spec-driven development, both open-source tooling the author maintains. Because the reasoning was documented rather than implicit, the frozen record could be subjected to adversarial review by an independent frontier model, which returned a blocked verdict with 11 findings, three of them critical, spanning identifiability, measurement error, selection, validation leakage, and compute. Each finding was resolved within the same recorded loop: the falsification gates and controls were frozen before any real result was inspected, the estimator was selected as in Section 3, and a mis-specified reliability aggregation (a product of coefficients) was replaced by the joint coefficient of Section 2.2. The resolutions were then cross-verified across models with a second adversarial pass and with `parallel-ai-agents` (Cheng, 2026b), a package the author maintains that dispatches a task to independent agents and cross-compares their outputs. The recorded decisions, the blocked verdict, and every resolution are versioned in the repository.

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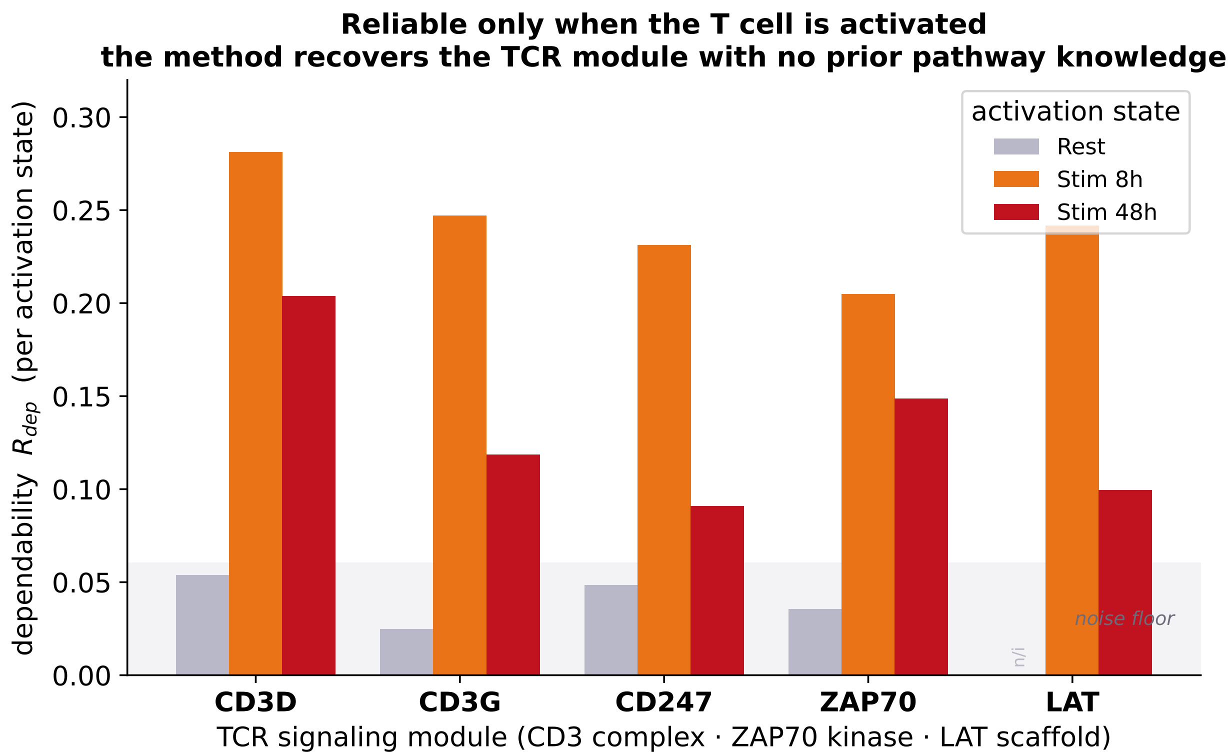

Figure 4: Per-state dependability for the T-cell-receptor signaling module. Resting values sit at the noise floor; the effect becomes reliably measurable only upon activation.

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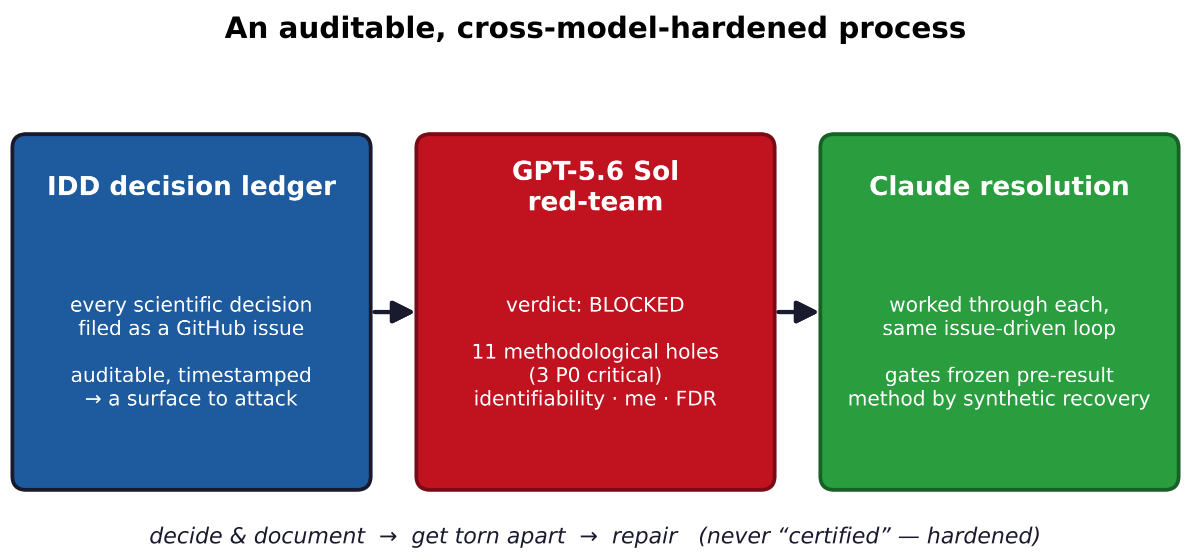

Figure 5: The recorded, cross-model-hardened process: a decision ledger enables adversarial review by an independent model, whose blocked verdict is resolved in the same loop. The verdict was blocked and worked through, not certified.

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